INCH-POUND

MIL-I-82931A(OS)
30 November 1994
SUPERSEDING
MIL-I-82931(OS)
21 June 1994

MILITARY SPECIFICATION

INFLATION DEVICE, FLU-8B/P (FW98) REMANUFACTURE SPECIFICATION

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the remanufacture requirements for reworking FLU-8A/P to the upgraded FLU-8B/P Inflation Device.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplements thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

PPP-B-636	Boxes, Shipping, Fiberboard

MILITARY

MIL-B-81705	Barrier Materials, Flexible, Electrostatic Protective, Heat Sealable

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Indian Head Division, Naval Surface Warfare Center, Standardization Branch (Code 8420), 101 Strauss Avenue, Indian Head, MD 20640-5035 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by sending a letter.

AMSC N/A FSC 1377

DISTRIBUTION STATEMENT A: Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:31Z
Check the source to verify that this is the current version before use.

STANDARDS

MILITARY

MIL-STD-109	Quality Assurance Terms and Definitions
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-453	Inspection, Radiographic
MIL-STD-1168	Ammunition Lot Numbering

(Unless otherwise specified, copies of federal and military specifications and standards are available from Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following Government drawings and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation (see 6.2).

DRAWINGS

NAVAL AIR SYSTEM COMMAND (30003)

849AS153	Inflation Device, Automatic, FLU-8B/P
849AS103	Battery, Dry Storage

(Application for copies should be addressed to: Commander, Indian Head Division, Naval Surface Warfare Center, Indian Head, MD 20640-5035.)

CODE OF FEDERAL REGULATIONS (CFR)

49 CFR 100-199	Transportation

(Application for copies of CFRs should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402-0001.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order (see 6.2), a sample shall be subjected to the first article inspection (see 4.4 and 6.4).

3.2 General. Materials, components, and assemblies shall comply with the requirements of Drawing 849AS153.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:31Z
Check the source to verify that this is the current version before use.

3.3 **New materials.** The following items shall replace parts discarded from the FLU-8A/P in the remanufacture of this inflation device. The parts are available from Conax Florida Corporation as Refurbishment Kit, FLU-8B/P Conax P/N 1512-173-01. The kit parts are packaged in an anti-static bag unless otherwise noted below.

ITEM	Part Number	QTY
Forward Locking Piercing Pin	1112-242-01	1 each
Sensor Housing Assembly	1813-044-02	1 each
Nameplate Left Hand Note 1	1216-084-01	1 each
Nameplate Right Hand Note 1	1216-084-02	1 each
Retainer Washer	1116-108-01	1 each
O-Ring (Primer)	1202-003-14	1 each
O-Ring (Piercing Pin)	1202-004-14	1 each
O-Ring (Piercing Pin)	1202-006-14	1 each
O-Ring (CO_2 Cylinder)	MS28775-012	1 each
Screw, Rd HD One-Way #5-40X, .31LG	CC-327-01	4 each
Washer	1116-085-01	1 each
Primer Plug	1102-181-01	1 each
Retainer Plug	1102-216-01	1 each
Warning Label	1216-056-01	1 each
Primer Note 2	CC-114-01	1 each
Lubricant Note 3	CC-209-02	A/R
Packing Loop	1138-003-01	1 each
Safety Thread	CC-361-01	A/R
Lubricant (for Oversized Indicator Pin) Note 4	Note 4	A/R
Indicator Pin (Oversized) Note 5	1131-059-02	A/R
Cylinder, Carbon Dioxide (35 grams)	CC-520-1	1 each
Loctite 290	CC-3022-01	A/R

Notes:

1. Nameplates are packaged separately on pin feed backing. One extra nameplate is included for every ten to allow for printing and application fall-out.

2. This item is packaged and shipped separately.

3. One container of lubricant is shipped for every 250 refurbishment kits.

4. This lubricant is not part of the refurbishment kit. The lubricant is made from 1 part Vydax IPA lubricant or equivalent and four parts alcohol. Vydax IPA is supplied under the name Nyefilm 501 D/IPA. A suggested manufacturer is W.F. Nye, PO Box G 927, New Bedford MA 02742.

5. Oversized indicator pins are used during rework only when a reworked items fails pressure test at the indicator pin port.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:31Z
Check the source to verify that this is the current version before use.

3.4 Reusable material.

ITEM	Part Number	QTY
Indicator Pin[Note 1]	1131-059-01	1 each
Body, Aid-Pak	1108-065-01	1 each
Bottom Gasket	1116-021-01	1 each
Top Gasket	1116-021-02	1 each
Cam Lever	1136-022-01	1 each
Pin, Ground Contact	CC-350-03	1 each
Pin, Grooved (Type 24, 1/8 Dia, 5/8 Long)	CC-350-04	1 each
Protective Plug	CC-355-11	1 each
Spring	MS24585-C102	1 each

Notes:

1. This part is replaced by the oversized indicator pin, p/n 1131-059-02, only if the reworked item fails the pressure test at the indicator pin port.

3.5 Primer (M101). The primer, Conax p/n CC-114, shall have been manufactured within 12 months of the FLU-8 manufacture date. The procedure and sample size for lot acceptance testing of the primer is the responsibility of the supplier. The lot acceptance procedure, however, must verify the following:

3.5.1 Visual examination (C1). One hundred percent visual examination shall be performed by the manufacturer. Each primer shall be visually inspected to verify that the material, construction, dimensions, and markings comply with the requirements of this specification. Each primer shall be inspected for presence of epoxy seal at pin end.

3.5.2 Bridgewire resistance (C2). One hundred percent bridgewire resistance shall be performed by the manufacturer. The bridgewire resistance of each primer shall be determined and recorded using a test meter having an accuracy of $\pm$ 0.1 ohms. Only primers having a bridgewire resistance of 3.5 $\pm$ 1.5 ohms shall be considered acceptable.

3.5.3 Radiographic examination (C3). Prior to assembly into the FLU-8, the primer shall be radiographically examined in accordance with MIL-STD-453. Each primer shall be X-rayed on boards or trays. The X-ray negatives shall be identified by item nomenclature and lot number. Primers shall be left mounted to the boards or trays until after the X-rays have been reviewed and defects identified. Defective items shall be removed from the production lot. After screening, the X-rays with defectives identified shall be forwarded to the engineering agency designated by the procuring activity. Film interpretation of the radiographs shall verify the following:

a. No material defects
b. Powder charges present, properly located, and not mixed.

3.5.4 Sample acceptance testing (nondestructive) (M102). The lot acceptance procedure for the primer shall include the following sampling dimensional inspections: body length, pin length, body outer diameter, pin outer diameter. Visual sampling inspection shall include the following: inspect

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:31Z
Check the source to verify that this is the current version before use.

for presence of red sealant at pin end, inspect packaging marking for legible and correct manufacturing dates and part number, inspect for evidence of poor workmanship.

3.5.5 Sample acceptance testing (destructive) (C4). The final lot acceptance procedure for the primer shall include the following sampling tests:

3.5.5.1 No-fire current. The primer shall not ignite or dud when bridgewire is subjected to 0.045 to 0.050 amperes for at least 30 seconds.

3.5.5.2 Performance testing. The primer shall fire upon application of current discharge of a 16 ± 10 percent microfarad capacitor charged to 2.5 volts. The primer shall be assembled and tested in the Conax energy monitor fixture (SK23061). Projectile penetration in the monitoring disc shall be 0.035 inch minimum and 0.095 inch maximum.

3.6 Subassembly testing. Testing is required to be performed on the sensor and body subassemblies by the contractor as noted in Drawing 849AS153.

3.6.1 Sensor assembly functional test (C5). Each sensor assembly (see view in 849AS153) shall be subjected to a simulated firing as stated in Drawing 849AS153. Any failure shall be cause for rejection of the sensor assembly.

3.6.2 Body assembly leakage tests (C6). Each body assembly (see view in Drawing 849AS153), prior to installing the primer, shall be subjected to nitrogen leak tests of Drawing 849AS153. Two tests are required for each body assembly, Leakage-A and Leakage-B.

3.6.2.1 Leakage-A. A body assembly failing this test shall be rejected. If desired, the indicator pin, Conax p/n 1139-059-01, can be replaced by the oversized indicator pin, Conax p/n 1139-059-02, and the Leakage-A test repeated. If the assembly fails a second time, the assembly shall be rejected and scrapped.

3.6.2.1 Leakage-B. A body assembly failing this test shall be rejected.

3.6.3 Manual operation test (C7). Prior to installation of the nylon safety thread, each body assembly shall be subject to the manual operation tests of Drawing 849AS153. Any body assembly failing any of the tests shall be rejected.

3.7 Bleed circuit resistance test (C8). Each assembled FLU-8B/P shall be subjected to the bleed circuit resistance tests of Drawing 849AS153.

3.8 Assembly tests (M103). First article and lot acceptance samples shall be subject to visual, radiographic, and battery voltage tests as stated in Drawing 849AS153.

3.9 Workmanship (M104). Workmanship shall ensure safety, proper orientation, and service life by making units that are free of the following; cracks or voids, foreign matter, missing or improperly assembled components, cracked, flaked, or insufficient surface coatings, material defects, illegible or incorrect markings, corrosion, or mechanical defects such as burrs, sharp edges and dents.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:31Z
Check the source to verify that this is the current version before use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the manufacturer is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the manufacturer may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements (see 6.2).

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements; however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Quality assurance terms and definitions. Reference shall be made to MIL-STD-109 to define the quality assurance terms used.

4.1.3 Inspections. Inspections shall be in accordance with MIL-A-2550 and MIL-STD-453.

4.1.4 Examination, tests, and facility. After provisional acceptance at source, first article or lot acceptance samples shall be inspected for all requirements of the drawings and specifications at a Government laboratory or other such facility specified in the contract (see 6.2).

4.2 Lot size. The production lot size shall be as specified in the contract (see 6.2). The maximum lot size shall be 1200, exclusive of test samples, and shall consist of primer assemblies from not more than one lot.

4.3 Examination. One hundred percent examination shall be performed for all non-destructive critical defects defined in this specification or Drawing 849AS153. Examination for all major and minor defects defined in this specification or Drawing 849AS153 shall be performed in accordance with the classification of defects. Four power (4x) magnification may be used for inspection of such defects as burrs and foreign matter. When compliance with the applicable requirements is in doubt as a result of visual examination, the characteristic shall be measured or gaged to determine acceptability. All nonconforming material shall be rejected.

4.4 First article testing. Unless otherwise specified in the contract, purchase order, or work order, manufacture and test of a first article sample is required when more than 18 months have elapsed since manufacture of the latest accepted FLU-8B/P production lot. First article testing shall be as specified in Drawing 849AS153. Failure of any FLU-8B/P to meet the requirements of this specification and Drawing 849AS153 shall be cause for rejection of the first article sample.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:31Z
Check the source to verify that this is the current version before use.

4.5 Lot acceptance testing. Lot acceptance testing shall be in accordance with Drawing 849AS153. Failure of any FLU-8B/P to meet the requirements of this specification and Drawing 849AS153 shall be cause for rejection of the lot represented.

5. PACKAGING

5.1 Preservation and packing (M105). In addition to the following, Level A and commercial preservation/packing shall be in accordance with 49 CFR 171-178.

5.1.1 Packaging for service use. Level A preservation and packing shall be used for packaging of all production lots for service use. For an explanation of terms, see 6.5.

5.1.1.1 Preservation (inner container). Each FLU-8 shall be individually packaged in a heat-sealed antistatic plastic bag conforming to MIL-B-81705, type II. The CO_2 cylinder shall not be installed on the assembly when packaged, but shall be packaged within the same bag in a separate heat-sealed compartment.

5.1.1.2 Packing (outer container). FLU-8 assemblies packaged in accordance with 5.1.1.1 shall be packed in a container in accordance with PPP-B-636. Internal bracing shall be such as to afford protection against physical damage or deformation to the FLU-8B/P assemblies, or the internal packaging. Unless otherwise specified in the contract or purchase order, the number of inner containers packed in the outer container shall be at the option of the supplier. Only FLU-8 assemblies having the same lot number shall be packed in a single outer container.

5.1.2 Packaging for immediate use. Commercial preservation and packing shall be used for packing FLU-8 assemblies which are intended for expenditure in preproduction and production lot acceptance testing. For explanation of terms, see 6.5.

5.2 Marking (M106).

5.2.1 Special marking. Marking of outer containers shall be in accordance with 49 CFR 171-178 and MIL-STD-129.

5.2.2 Normal marking. Unless otherwise specified in the contract or purchase order, the marking information of the inner and outer containers shall be as specified below. The specified markings shall be applied to the containers in accordance with applicable provisions of MIL-STD-129.

Inner Container Marking

a. NSN 1377-01-364-4089/DODIC FW98
b. INFLATION DEVICE, AUTOMATIC, FLU-8B/P
c. 30003-849AS153
d. 1 each
e. Lot number in accordance with MIL-STD-1168
f. Serial Number

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:31Z
Check the source to verify that this is the current version before use.

Outer Container Marking

a. NSN 1377-01-364-4089/DODIC FW98
b. INFLATION DEVICE, AUTOMATIC, FLU-8B/P
c. 30003-849AS153
d. Quantity of individual FLU-8B/P assemblies
e. Gross weight and cube
f. Lot number in accordance with MIL-STD-1168
g. Serial Numbers
h. Contract/Purchase Order number
i. UN SER. NO.:0323
j. PSN: Cartridges, Power Device
k. Haz Classification 1.4S
l. POP Markings

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory).

6.1 Intended use. The automatic inflation device provides power to automatically inflate the aircrew's life vest after entry into seawater. This specification is for reworking FLU-8A/P, Drawing 849AS150, to the upgraded configuration FLU-8B/P, Drawing 849AS153. Conax document IM-69, revision C, contains approved rework procedures. Conax document QAP-23, original version, contains an approved quality assurance plan for acceptance testing for CC-114-01 Primer.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, date, and revision letter of this specification.

b. Issue of DODISS to be cited in the solicitation and, if required, the specific issue of individual documents referenced and that MIL-STD-810 shall be used.

c. Whether a first article sample is required (see 4.4 and 6.4).

d. Test facility for first article tests (see 4.1).

e. Production test activity and lot size (see 4.1 and 4.2).

f. Any special test conditions (see 4.3).

g. Directions for shipping radiographs of entire lot along with ballistic sample to the activity conducting production lot acceptance tests (see 3.5.3).

h. Level of packing and marking.

i. Ammunition data cards in accordance with MIL-STD-1167.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:31Z
Check the source to verify that this is the current version before use.

j. That the safety precaution requirements of the "Contractor's Safety Manual for Ammunition, Explosives, and Related Dangerous Material," DOD 4145.26M are applicable. NOTE: When this specification is used as part of the description of work to be accomplished by a Government activity, the safety precaution requirements of "Ammunition and Explosives Ashore" OP 5 are applicable.

k. Submission of rework procedures, CC-114 acceptance test and inspection procedures, and CC-114 Primer acceptance test and inspection results.

l. Certification of conformance that all deliverables met requirements of section 3 of this specification.

6.3 Hazard notice. The inflation device described herein and certain of its components are flammable or explosive and present hazards in manufacture, handling, storage, and shipment. The contractor should recognize these properties and take appropriate measures to guard and protect against fire, explosion, adverse environment, corrosive atmosphere, rough handling, and electrically induced incidents.

6.4 First article. When a first article inspection is required, the contracting officer should provide specific guidance to offerors whether the items should be a first article sample or a sample selected from the first production items (see 4.4), and the number of items to be tested as specified in 4.4. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examination, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.5 Explanation of terms. For the purpose of this specification, NAVSUPINST 4030.28, when applicable, shall be used to clarify terms.

6.6 Subject term (keyword) listing.

Aircrew
Electric actuation
Life vest

Preparing Activity:
Navy - OS
(Project1377-NE87)

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:31Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-I-82931A(OS)	941130

3. DOCUMENT TITLE
INFLATION DEVICE, FLU-8B/P (FW98) REMANUFACTURE SPECIFICATION

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)	
COMMANDER, INDIAN HEAD DIVISION NAVAL SURFACE WARFARE CENTER	(1) Commercial 301-743-4358/4510	(2) AUTOVON 354-4358/4510
c. ADDRESS (Include Zip Code) 101 STRAUSS AVE., (CODE 8420) INDIAN HEAD, MD 20640-5035	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

DD Form 1426, OCT 89 Previous editions are obsolete. 198-290

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:31Z
Check the source to verify that this is the current version before use.